

PRAKTIKUM PEKAN 10 PEMROGRAMAN WEB
JAVASCRIPT LANJUTAN



Telkom
University

FUTUWAH IQAMATUL HAQI

707012500099

49-04

1. Tujuan Praktikum

- Memahami konsep pembuatan aplikasi web menggunakan Vanilla JavaScript (tanpa framework).
- Menerapkan konsep State Management sederhana.
- Mengimplementasikan fitur interaktif seperti Add to Cart, Cart Management, dan Dynamic Rendering.
- Memahami manipulasi DOM, event handling, serta modularisasi kode JavaScript.
- Mempelajari struktur folder dan best practices dalam pengembangan front-end vanilla.

2. Alat & Bahan

- Visual Studio Code / Text Editor
- Browser (Google Chrome / Firefox)
- Laptop

3. Langkah-Langkah

Praktikum dimulai dengan persiapan struktur folder proyek yang terdiri dari direktori core, components, api, dan utils. Selanjutnya dibuat sistem state management sederhana pada file Store.js yang dilengkapi dengan fungsi getState() dan setState(). Kemudian, dibuat fungsi deepClone() pada file utils.js untuk menghindari masalah reference object. Setelah itu, dikembangkan tiga komponen utama yaitu ProductList.js untuk menampilkan daftar produk secara dinamis, Cart.js untuk mengelola keranjang belanja beserta perhitungan total, dan Newsletter.js untuk form subscription. Pada file app.js, diintegrasikan seluruh komponen, ditambahkan fungsi addToCart dan checkout, serta dilakukan rendering ulang setiap kali state berubah. Selanjutnya dilakukan styling dengan CSS untuk membuat tampilan yang responsif dan menarik, termasuk efek hover dan animasi sederhana. Tahap terakhir adalah pengujian fungsionalitas aplikasi seperti menambahkan produk ke keranjang, memperbarui total harga, dan proses checkout.

4. Dokumentasi

Kode

Index.html

```
<!doctype html>
<html lang="id">
<head>
<meta charset="UTF-8" />
<title>Vanilla JS Web Store</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
```

```

<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Merriweather:ital,wght@0,400;0,700;1,400&display=swap" rel="stylesheet">
<link rel="stylesheet" href="style.css" />

<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
<script type="module" src="app.js" defer></script>
</head>

<body>
<div id="app">
<div class="container">

<div class="listing-header">
<nav class="breadcrumb" aria-label="Breadcrumb">
<a href="#">
<svg xmlns="http://www.w3.org/2000/svg" fill="currentColor" viewBox="0 0 20 20">
<path d="m19.707 9.293-2-2-7a1 1 0 0 0-1.414 0l-7 7-2 2a1 1 0 0 0 1.414 1.414L2 10.414V18a2 2 0 0 0 2 2h3a1 1 0 0 0 1 1v-4a1 1 0 0 1 1-1h2a1 1 0 0 1 1 1v4a1 1 0 0 1 1 1h3a2 2 0 0 0 2-2v-7.586l.293.293a1 1 0 0 0 1.414 1.414Z" />
</svg>
Home
</a>
<svg style="color: #9CA3AF;" xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="m9 5 7 7-7 7" />

</svg>
<a href="#">Products</a>
<svg style="color: #9CA3AF;" xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="m9 5 7 7-7 7" />

```

```

</svg>
<span style="color: var(--text-muted);">Electronics</span>
</nav>

<div class="toolbar">
<h2 class="page-title">Electronics</h2>
<div class="toolbar-actions">
<button class="btn-secondary">

<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
<path stroke="currentColor" stroke-linecap="round" stroke-width="2" d="M18.796
4H5.204a1 1 0 0 0-.753 1.659l5.302 6.058a1 1 0 0 1 .247.659v4.874a.5.5 0 0 0 .243
2.25a.5.5
0 0 0 .8-.4v-7.124a1 1 0 0 1 .247-.659l5.302-6.059c.566-.646.106-1.658-.753-1.658Z" />
</svg>
Filters
</button>
<button class="btn-secondary">
<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-
width="2" d="M7 4v16M7 4l3 3M7 4 4 7m9-3h6l-6 6h6m-6.5 10 3.5-7 3.5 7M14 18h4"
/>

</svg>
Sort
</button>
</div>
</div>
</div>

<div id="product-list" class="product-grid"></div>

<div id="cart" class="cart-section"></div>
</div>
</div>

<script type="module" src="app.js"></script>
</body>
</html>

```

Api/productApi.js

```
export const fetchProducts = async () => {
```

```
return new Promise((resolve) => {

  setTimeout(() => {
    resolve([
      {
        id: 1,
        title: "Mechanical Keyboard",
        price: 150,
        rating: { rate: 4.8, count: 230 },

        image: "https://images.unsplash.com/photo-1618384887929-16ec33fab9ef?auto=format&fit=crop&w=800&q=80"
      },
      {
        id: 2,
        title: "Gaming Mouse",
        price: 80,
        rating: { rate: 4.6, count: 150 },

        image: "https://images.unsplash.com/photo-1527814050087-3793815479db?auto=format&fit=crop&w=800&q=80"
      },
      {
        id: 3,
        title: "4K Monitor 24\"",
        price: 300,
        rating: { rate: 4.9, count: 85 },

        image: "https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?auto=format&fit=crop&w=800&q=80"
      },
      {
        id: 4,
        title: "Premium Coffee Beans",
        price: 25,
        rating: { rate: 4.7, count: 500 },

        image: "https://images.unsplash.com/photo-1559056199-641a0ac8b55e?auto=format&fit=crop&w=800&q=80"
      },
      {
        id: 5,
```

```

title: "Travel Tumbler",
price: 35,
rating: { rate: 4.5, count: 120 },

image: "https://images.unsplash.com/photo-1578500494198-
246f612d3b3d?auto=format&fit=crop&w=800&q=80"
},
{
id: 6,
title: "Noise Cancelling Headphones",
price: 200,
rating: { rate: 4.8, count: 320 },

image: "https://images.unsplash.com/photo-1505740420928-
5e560c06d30e?auto=format&fit=crop&w=800&q=80"
},
{
id: 3,
title: "Brown T-Shirt",
price: 75,
rating: { rate: 4.9, count: 256 },

image: "https://69fc19f4388c948cecb8f7fb.imgix.net/T-shirt.JPG?w=1024&h=1536"
},
{
id: 3,
title: "Leather Bag",
price: 60,
rating: { rate: 4.9, count: 108 },

image:
"https://69fc19f4388c948cecb8f7fb.imgix.net/leather%20bag.JPG?w=570&h=711"
},
]);
}, 800);
});
};

```

Components/Cart.js

```

export function Cart(cart, onCheckout) {
const container = document.getElementById("cart");
container.innerHTML = "<h2>Your Cart</h2>";

```

```

if (cart.length === 0) {
  container.innerHTML += "<p class='empty-msg'>Cart is empty.</p>";
  return;
}

const list = document.createElement("div");
list.className = "cart-list";

cart.map((item) => {
  const div = document.createElement("div");
  div.className = "cart-item";
  div.innerHTML = `

<div class="cart-details">
<h4>${item.title}</h4>
<span>${item.price}</span>
</div>
`;
  list.appendChild(div);
});

container.appendChild(list);

const total = cart.reduce((acc, item) => acc + (item.price || 0), 0);

const footer = document.createElement("div");
footer.className = "cart-footer-summary";
footer.innerHTML = `
<div class="total-row">
<span>Total:</span>
<span class="total-price">${total.toFixed(2)}</span>
</div>
<button class="btn-primary" id="checkout-btn">Buy Now</button>
`;

footer.querySelector("#checkout-btn").addEventListener("click", onCheckout);
container.appendChild(footer);
}

```

Components/Newsletter.js

```

export function Newsletter() {
  const $container = $('<div class="newsletter-section"></div>');
  const $content = $(

```

```

<div class="newsletter-content">
<h3>Subscribe to our updates</h3>
<p>Get the latest tech news and offers.</p>
<form id="newsletter-form">
<input type="email" placeholder="Enter your email" required />
<button type="submit" class="btn-primary">Subscribe</button>
</form>
<div class="success-message" style="display:none; margin-top: 10px; color: green;">

Thanks for subscribing!
</div>
</div>
`);

$content.append($content);
$("#app").append($container);

$content.find("form").on("submit", function (e) {
E.preventDefault();
const $form = $(this);

$form.slideUp(400, function() {
// Show success message with fade in
$content.find(".success-message").fadeIn();
});
});
}

```

Components/ProductList.js

```

export function ProductList(products, onAdd) {
const container = document.getElementById("product-list");
container.innerHTML = "";

products.map((product) => {
const card = document.createElement("div");
card.className = "product-card";

const discount = Math.floor(Math.random() * 20) + 5;
const rating = product.rating || { rate: 4.5, count: 100 };

card.innerHTML = `
<div class="card-image">


```



```

</div>
<div class="card-content">
<span class="badge">Up to ${discount}% off</span>
<a href="#" class="card-title" title="${product.title}">${product.title}</a>

<div class="rating-row">
<div class="stars">
${getStarSVG().repeat(5)}
</div>
<span class="rating-score">${rating.rate}</span>
<span class="rating-count">(${rating.count})</span>
</div>

<ul class="features-list">
<li class="feature">
<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-
width="2" d="M13 7h6l2 4m-8-4v8m0-8V6a1 1 0 0 0-1-1H4a1 1 0 0 0-1 1v9h2m8
0H9m4

0h2m4 0h2v-4m0 0h-5m3.5 5.5a2.5 2.5 0 1 1-5 0 2.5 2.5 0 0 1 5 0Zm-10 0a2.5 2.5 0 1 1-
5 0
2.5 2.5 0 0 1 5 0Z" />
</svg>
Fast Delivery
</li>
</ul>

<div class="card-footer">
<span class="price">${product.price}</span>
<button class="add-btn btn-primary">
<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-
width="2" d="M4 4h1.5L8 16m0 0h8m-8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-4Zm8 0a2 2 0 1 0 0
4 2 2

0 0 0 0-4Zm.75-3H7.5M11 7H6.312M17 4v6m-3-3h6" />
</svg>
Add
</button>
</div>

```

```

</div>
`;

card.querySelector(".add-btn").addEventListener("click", function () {
  onAdd(product);
});

container.appendChild(card);
});
}

function getStarSVG() {
  return `
<svg xmlns="http://www.w3.org/2000/svg"
fill="currentColor"
viewBox="0 0 20 20"
width="20"
height="20">
<path d="M9.049 2.927c.3-.921 1.603-.921 1.902 0l1.07 3.292a1
1 0 00.956h3.462c.969 0 1.371 1.24588 1.81l-2.8 2.034a1
1 0 00-.364 1.118l1.07 3.292c.3.921-.755 1.688-1.54
1.118l-2.8-2.034a1 1 0 00-1.175 0l-2.8
2.034c-.784.57-1.838-.197-1.539-1.118l1.07-3.292a1
1 0 00-.364-1.118L2.98 8.72c-.783-.57-.38-1.81.588-1.81h3.461a1
1 0 00.951-.69l1.07-3.292z"/>
</svg>`;
}

```

Core/Store.js

```

import { deepClone } from "../utils.js";

let state = {
  products: [],
  cart: [],
};

export const getState = () => deepClone(state);

export const setState = (newState) => {

  const nextState = { ...state, ...newState };

  state = nextState;
}

```

```
console.log("[Store] State updated:", state);  
};
```

Core/utils.js

```
import { getState } from "../Store.js";  
  
export function deepClone(obj) {  
  if (obj === null || typeof obj !== "object") {  
    return obj;  
  }  
  
  if (obj instanceof Date) {  
    return new Date(obj.getTime());  
  }  
  
  if (Array.isArray(obj)) {  
    return obj.map((item) => deepClone(item));  
  }  
  
  const clonedObj = {};  
  for (const key in obj) {  
    if (Object.prototype.hasOwnProperty.call(obj, key)) {  
      clonedObj[key] = deepClone(obj[key]);  
    }  
  }  
  
  return clonedObj;  
}
```

App.js

```
import { fetchProducts } from "../api/productApi.js";  
import { getState, setState } from "../core/Store.js";  
import { ProductList } from "../components/ProductList.js";  
import { Cart } from "../components/Cart.js";  
  
import { Newsletter } from "../components/Newsletter.js";  
  
function addToCart(product) {  
  const state = getState();  
  
  setState({
```

```

    cart: [...state.cart, product],
  });

  render();
}

function checkout() {
  setState({ cart: [] });
  alert("Thank you for your purchase!");
  render();
}

function render() {
  const state = getState();

  ProductList(state.products, addToCart);
  Cart(state.cart, checkout);
}

async function init() {
  try {
    const data = await fetchProducts();
    setState({ products: data });
    render();
    Newsletter();
  } catch (error) {
    console.error("Failed to load products:", error);
  }
}

init();

```

Style.css

```

:root {

  --primary: #0D9488; --primary-hover: #0F766E; --primary-light: #CCFBF1;
  --bg-color: #F9FAFB; --card-bg: #FFFFFF;
  --text-main: #111827; --text-muted: #6B7280;
  --border-color: #E5E7EB; --rating-star: #FACC15;
  --font-serif: 'Merriweather', serif; --font-sans: 'Inter', sans-serif;
  --radius: 8px;
}

```

```
* { box-sizing: border-box; margin: 0; padding: 0; }

body { font-family: var(--font-sans); background: var(--bg-color); color: var(--text-main); line-height: 1.5; -webkit-font-smoothing: antialiased; }

.container { max-width: 1280px; margin: 0 auto; padding: 32px 16px; }

.flex-row, .breadcrumb, .breadcrumb a, .toolbar, .toolbar-actions, .feature, .rating-row, .card-
footer { display: flex; align-items: center; }

.btn-primary, .btn-secondary { display: inline-flex; align-items: center; justify-content: center;
gap: 8px; border-radius: var(--radius); font-size: 0.875rem; font-weight: 500; cursor: pointer;
transition: 0.2s; }
svg { width: 16px; height: 16px; }

.listing-header { margin-bottom: 32px; }
.page-title { font-family: var(--font-serif); font-size: 1.875rem; font-weight: 700; }
.toolbar { justify-content: space-between; flex-wrap: wrap; margin-top: 16px; gap: 16px; }
.toolbar-actions { gap: 12px; }

.breadcrumb { gap: 8px; font-size: 0.875rem; color: var(--text-muted); margin-bottom: 12px; }
.breadcrumb a { color: var(--text-muted); text-decoration: none; gap: 6px; }
.breadcrumb a:hover { color: var(--primary); }

.btn-secondary { background: white; border: 1px solid var(--border-color); color: var(--text-main); padding: 8px 12px; }

.btn-secondary:hover { background: #F3F4F6; color: var(--primary); }

.btn-primary { background: var(--primary); color: white; border: none; padding: 10px 20px; }
.btn-primary:hover { background: var(--primary-hover); }
.btn-primary svg { width: 20px; height: 20px; }

.product-grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));
gap: 24px; margin-bottom: 32px; }
```

```

.product-card { background: var(--card-bg); border: 1px solid var(--border-color);
border-radius: var(--radius); overflow: hidden; display: flex; flex-direction: column;
transition: box-
shadow 0.2s; box-shadow: 0 1px 2px 0 rgba(0,0,0,0.05); }

.product-card:hover { box-shadow: 0 4px 6px -1px rgba(0,0,0,0.1); }

.card-image { height: 220px; padding: 24px; display: flex; justify-content: center;
background:
white; }
.card-image img { max-height: 100%; max-width: 100%; object-fit: contain; }
.card-content { padding: 24px; display: flex; flex-direction: column; flex: 1; }

.badge { background: var(--primary-light); color: var(--primary); font-size: 0.75rem;
font-weight: 600; padding: 2px 10px; border-radius: 4px; width: fit-content; margin-
bottom: 12px;
}
.card-title { font-size: 1.125rem; font-weight: 600; color: var(--text-main); text-
decoration:
none; margin-bottom: 8px; line-height: 1.4; display: -webkit-box; -webkit-line-clamp: 2;
line-clamp: 2; -webkit-box-orient: vertical; overflow: hidden; }

.card-title:hover { text-decoration: underline; }

.rating-row { gap: 8px; margin-bottom: 16px; }
.stars { display: flex; color: var(--rating-star); }
.rating-score { font-size: 0.875rem; font-weight: 600; }
.rating-count, .feature { font-size: 0.875rem; color: var(--text-muted); }
.features-list { display: flex; gap: 12px; margin-bottom: 20px; }
.feature { gap: 6px; }
.card-footer { margin-top: auto; justify-content: space-between; }
.price { font-size: 1.5rem; font-weight: 800; }

#cart-sidebar { background: white; border-left: 1px solid var(--border-color); padding:
24px;
width: 350px; position: fixed; right: 0; top: 0; bottom: 0; box-shadow: -4px 0 16px
rgba(0,0,0,0.05); overflow-y: auto; transform: translateX(100%); transition: transform
0.3s; z-index: 50; display: flex; flex-direction: column; }

```

```

#cart-sidebar.open { transform: translateX(0); }
#cart-sidebar h2 { margin-bottom: 24px; font-family: var(--font-serif); }
.empty-msg { color: var(--text-muted); font-style: italic; }

.cart-list { flex: 1; overflow-y: auto; display: flex; flex-direction: column; gap: 16px;
margin-bottom: 24px; }

.cart-item { display: flex; align-items: center; gap: 12px; padding-bottom: 12px; border-
bottom: 1px solid var(--border-color); }

.cart-item img { width: 50px; height: 50px; object-fit: cover; border-radius: 4px; border:
1px
solid var(--border-color); }

.cart-details h4 { font-size: 0.875rem; font-weight: 600; margin-bottom: 4px; display: -
webkit-
box; -webkit-line-clamp: 1; line-clamp: 1; -webkit-box-orient: vertical; overflow: hidden;
}

.cart-details span { font-size: 0.875rem; color: var(--text-muted); }

.cart-footer-summary { border-top: 2px solid var(--border-color); padding-top: 20px;
margin-top: auto; }

.total-row { display: flex; justify-content: space-between; align-items: center; margin-
bottom:
16px; font-weight: 700; font-size: 1.125rem; }
.total-price { color: var(--primary); }
#checkout-btn { width: 100%; }

.newsletter-section { margin-top: 64px; background: var(--card-bg); border-top: 1px
solid var(--
border-color); padding: 48px 0; text-align: center; }
.newsletter-content { max-width: 500px; margin: 0 auto; }
.newsletter-content h3 { font-family: var(--font-serif); font-size: 1.5rem; margin-bottom:
8px; }
.newsletter-content p { color: var(--text-muted); margin-bottom: 24px; }
.newsletter-content form { display: flex; gap: 8px; }

.newsletter-content input { flex: 1; padding: 10px; border: 1px solid var(--border-color);
border-radius: var(--radius); }
#app { width: 100%; }

```

5. Analisis & Pembahasan

Praktikum ini berhasil mengimplementasikan sebuah web store sederhana menggunakan Vanilla JavaScript. State management manual dengan deepClone berhasil menjaga konsistensi data antar komponen. Modularisasi kode pada file Store.js, ProductList.js, Cart.js, dan Newsletter.js membuat program lebih terstruktur dan mudah dipelihara. Fitur add to cart, perhitungan total harga, serta rendering DOM berjalan dengan baik. Namun, aplikasi masih memiliki kekurangan seperti belum adanya fitur persistence (LocalStorage) dan fetch data produk yang masih simulasi.

6. Kesimpulan

Praktikum ini telah mencapai tujuannya dengan baik. Melalui proyek web store ini, saya berhasil memahami konsep penting Vanilla JavaScript seperti manipulasi DOM, event handling, dan state management sederhana. Aplikasi yang dibuat berfungsi dengan baik untuk menampilkan produk dan mengelola keranjang belanja. Pengalaman ini menjadi fondasi yang kuat untuk mempelajari front-end development lebih lanjut.

7. Lampiran

Link repositori GitHub

<https://github.com/AriqAhmadNayaka/WebPro4904/tree/FutuwahlqamatulHaqi-707012500099>